1. List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

Feature	IPv4	IPv6
Address Size	**32-bit**(Short)	**128-bit** (Very long)
Look / Appearance	**Numbers** separated by dots (e.g., 192.168.1.1)	**Numbers and letters** separated by colons (e.g., 2001:db8::1)
Total Addresses	**4.3 billion** (We have run out of them)	**340 undecillion** (Virtually infinite; will never run out)
Security	**Optional** (Needs extra setup)	**Built-in** (Designed to be secure by default)

2. Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

Here is a simple setup assigning unique static IPv4 addresses to the 4 home devices using the **192.168.10.0/24** network:

1. Laptop

- **IP Address:** 192.168.10.10
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.10.1

2. Phone

- **IP Address:** 192.168.10.11
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.10.1

3. Smart TV

- **IP Address:** 192.168.10.12
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.10.1

4. Printer

- **IP Address:** 192.168.10.13
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.10.1

3. Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

Using a **/27 CIDR subnet mask** on a 192.168.1.0 network creates 8 valid subnets with 30 usable hosts per subnet.

The Calculation

- **Subnet Mask:** A /27 mask means the first 27 bits are locked for the network (255.255.255.224).
- **Number of Subnets:** Since a standard Class C network starts at /24, we borrowed 3 bits (27 - 24 = 3). (2^3) = **8 valid subnets**.
- **Hosts per Subnet:** We have 5 bits left for devices (32 - 27 = 5). Total size is (2^5 = 32). Subtracting 2 addresses (one for the network ID, one for the broadcast) leaves **30 usable host IPs**. The subnets increment by 32.

First Three Subnet Ranges

Subnet 1

- **Network Address:** 192.168.1.0
- **Usable Host Range:** 192.168.1.1 to 192.168.1.30
- **Broadcast Address:** 192.168.1.31

Subnet 2

- **Network Address:** 192.168.1.32
- **Usable Host Range:** 192.168.1.33 to 192.168.1.62
- **Broadcast Address:** 192.168.1.63

Subnet 3

- **Network Address:** 192.168.1.64
- **Usable Host Range:** 192.168.1.65 to 192.168.1.94
- **Broadcast Address:** 192.168.1.95

4. You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Hint: Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.

Using the bitwise AND method, the network details for the IP address **10.0.5.17** with a subnet mask of **255.255.255.0** are calculated below.

Step-by-Step Calculation

1. Convert to CIDR Notation

- A subnet mask of 255.255.255.0 means the first three octets are completely full of 1s in binary (8 + 8 + 8 = 24).
- In CIDR notation, this is a **/24** network.

2. Bitwise AND for Network Address

To find the network address, we align the IP address and the subnet mask in binary and perform a logical AND operation (where 1 AND 1 = 1, and anything else is 0 :

- **IP Address (10.0.5.17):** 00001010.00000000.00000101.00010001
- **Subnet Mask (255.255.255.0):** 11111111.11111111.11111111.00000000
- **Bitwise AND Result:** 00001010.00000000.00000101.00000000
- Converting the result back to decimal gives the network address: **10.0.5.0**

3. Determine Broadcast and Host Range

- Because it is a /24 network, the first three numbers (10.0.5.) are locked. Only the final number changes.
- The host bits are set to all 0s for the network address (.0) and all 1s for the broadcast address (.255).

Final Subnet Details

- **Network Address:** 10.0.5.0
- **Broadcast Address:** 10.0.5.255
- **Range of Valid Host Addresses:** 10.0.5.1 to 10.0.5.254

5. Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

